

Evidence for a role of human blood-borne factors in mediating age-associated changes in molecular circadian rhythms

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Abstract

Aging is associated with a number of physiologic changes including perturbed circadian rhythms; however, mechanisms by which rhythms are altered remain unknown. To test the idea that circulating factors mediate age-dependent changes in peripheral rhythms, we compared the ability of human serum from young and old individuals to synchronize circadian rhythms in culture. We collected blood from apparently healthy young (age 25-30) and old (age 70-76) individuals at 14:00 and used the serum to synchronize cultured fibroblasts. We found that young and old sera are equally competent at initiating robust ~24h oscillations of a luciferase reporter driven by clock gene promoter. However, cyclic gene expression is affected, such that young and old sera promote cycling of different sets of genes. Genes that lose rhythmicity with old serum entrainment are associated with oxidative phosphorylation and Alzheimer's Disease as identified by STRING and IPA analyses. Conversely, the expression of cycling genes associated with cholesterol biosynthesis increased in the cells entrained with old serum. Genes involved in the cell cycle and transcription/translation remain rhythmic in both conditions. We did not observe a global difference in the distribution of phase between groups, but found that peak expression of several clock-controlled genes (*PER3*, *NR1D1*, *NR1D2*, *CRY1*, *CRY2*, and *TEF*) lagged in the cells synchronized *ex vivo* with old serum. Taken together, these findings demonstrate that age-dependent blood-borne factors affect circadian rhythms in peripheral cells and have the potential to impact health and disease via maintaining or disrupting rhythms respectively.

eLife Assessment

The authors tested the hypothesis that age-dependent factors in human sera affect the core circadian clock or its outputs in cultured fibroblasts, and they provide **compelling** evidence that genes involved in the cell cycle and transcription/translation remain rhythmic in both conditions, genes associated with oxidative phosphorylation and Alzheimer's Disease lose rhythmicity in the aged condition, while the expression of cycling genes associated with cholesterol biosynthesis increase in the cells entrained with old serum. Together, the findings suggest that yet to be identified age-dependent blood-borne factors affect circadian rhythms in the periphery. The paper provides **fundamental** insights and a possible explanation for previous observations showing that circadian gene expression in peripheral tissues tends to dampen or phase-shift with age.

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Introduction

Circadian rhythms are known to regulate homeostatic physiology including sleep:wake, hormone production and body temperature, and their dysregulation with aging is accompanied by adverse health consequences^{1,2,3}, raising the possibility that health decline with age is caused in part by circadian dysfunction. Although the mechanisms responsible for age effects on circadian rhythms are unknown, signals from the central clock in the suprachiasmatic nucleus (SCN) dampen with age^{3,4,5} and rhythms change in peripheral tissues in different ways^{5,6}. Here we aimed to develop a cell culture model to study the effect of aging on human rhythms of peripheral tissues. Given that serum can reset the clock in peripheral fibroblasts⁷, we questioned the extent to which serum factors normally contribute to peripheral rhythms of gene expression and how they might affect rhythms with age, given that blood-borne factors can influence other aspects of aging⁸.

Using an established clinical study paradigm that sampled blood at 14:00⁹, we collected blood from young (age 25-30) and old (age 70-76) apparently healthy individuals with behavioral and physiological outputs quantified by wearable devices, and we tested the hypothesis that age-dependent factors in the sera affect circadian rhythms in cultured fibroblasts. In support of this hypothesis, we show here that genes associated with oxidative phosphorylation and mitochondrial functions lose rhythmicity in fibroblasts exposed to aged serum factors. We also find evidence of reduced entrainment in terms of altered expression of several molecular clock genes (*PER3*, *NR1D1*, *NR1D2*, *CRY1*, *CRY2* and *TEF*) when synchronized with aged serum. These findings suggest that age-related changes in blood borne factors contribute to impaired circadian physiology and the associated disease risks.

Results

We enrolled 8 old and 7 young human subjects (**Fig. 1A**, Tables S1 and **Fig. S2**), whose demographics are listed in **Table S1**. The behavioral and physiological assessments confirmed entrainment to the light-dark conditions local to the East Coast of the US. This is, for example, evident in the diurnal rhythms observed for physical activity (**Fig. 1C** top, vector magnitude two-way ANOVA for time-of-day $q=3.3E-06$), light exposure (lux two-way ANOVA for time-of-day

$q=0.01$), heart rate (bpm two-way ANOVA for time-of-day $q=0.016$) and sympathetic and parasympathetic nervous system indices derived from the Kubios heart rate variability analysis (two-way ANOVA for time-of-day $q=0.099$ and $q=0.063$, respectively).

Age-specific differences emerged for several outputs. The standard deviation of instantaneous heart rate values was significantly lower in the old compared to young (two-way ANOVA for age group $q=0.042$). As expected, the peak clock time (acrophase) of physical activity (triaxial accelerometry integrated as vector magnitude) was phase-advanced among old compared to young subjects, as was the midsleep time calculated from the MCTQ (Fig S1B), though not at a statistically significant level ($p=0.055$). On average, a lower rhythm-adjusted mean (mesor) heart rate of 55.8 ± 3.5 bpm was found in old compared to 69.7 ± 5.0 bpm in young, along with a higher heart rate variability (RR intervals) of 1106.7 ± 80.5 ms compared to 893.2 ± 61.8 ms, respectively (Fig 1B). Old subjects displayed lower activity in the sympathetic nervous system (SNS), and higher activity in the parasympathetic nervous system (PNS) compared to young (Fig. S1A), with no significant difference in cortisol levels (Fig. S1C). The overall high degree of variability rendered the cardiovascular trends, however, statistically not significant.

As noted above, blood was collected from these old/young individuals and the serum was used to synchronize BJ-5TA fibroblasts stably transfected with a *BMAL1-luciferase* construct^{7,9,11}. Circadian rhythms were assessed by luciferase assay¹⁰ over 4 days (Fig 2A). We conducted this study in a fibroblast line in order to build upon age-related changes found in a previous study which used human fibroblast lines⁹. An additional rationale for using this line is that it derives from normal human tissue, providing an advantage in studying normal physiology when compared to the commonly used U2OS line which is a genomically unstable osteosarcoma with chromosomal abnormalities¹². No significant differences were observed in the period, amplitude, and phase of the *BMAL1-luciferase* rhythm with young versus old serum treatment (Fig S3).

To assess serum effects on circadian gene expression, we first performed RNA-seq on fibroblasts synchronized with serum from a single old or single young individual at two hour intervals and found that, compared to the first day of synchronization, the second day (36-58 hours after synchronization) showed greater differences in MESORs between young and old serum-treated groups (Fig. S4). This is not surprising because the first day includes acute responses of the fibroblasts to serum, which can mask circadian rhythms, and so the second day is expected to reveal differences in endogenous rhythms between samples. Day 2 also revealed different phases of cyclic expression between young and old subjects for a larger number of genes. We proceeded to collect fibroblasts synchronized by sera from eight different subjects (four old, four young with two male and two female per group) at two-hour intervals, from 32 to 58 hours post serum addition. This added an extra two timepoints to the second day to facilitate the calculation of rhythmicity. Using CircaCompare¹³ and a weighted Bayesian Information Criterion (BIC) >0.75 cutoff, a significant number of genes were found to lose (1519 genes) or gain (637 genes) rhythmicity with age, underscoring the impact of age on the ability of serum to synchronize circadian rhythms, while 1209 genes were rhythmic in both groups (Fig 2B, 3A). Additionally, we used CircaCompare to estimate MESORs, amplitudes, and phases of gene oscillatory patterns in young and old groups, and to compare these cosinor parameters between groups (Fig 2C-D). Of the genes that were rhythmic in both groups, many also showed changes with age. For instance, 568 cyclically expressed genes showed a change in MESOR with age ($q < 0.05$ for MESOR differences), with MESOR values increasing for 163 genes and decreasing for 405 genes in the old serum-treated samples (Fig 2C, S5, right). Using q -values provided by CircaCompare we were able to detect changes in amplitude (Fig 2D, S5, left) for only a small number of genes (39 genes had decreased amplitude in old and 2 had increased amplitude). Using CircaCompare and a $q < 0.05$ cutoff for phase differences in genes rhythmic in young and old, we detected 20 genes with advanced phase, and 34 with delayed phases (Fig 2E). However, it is important to note that in

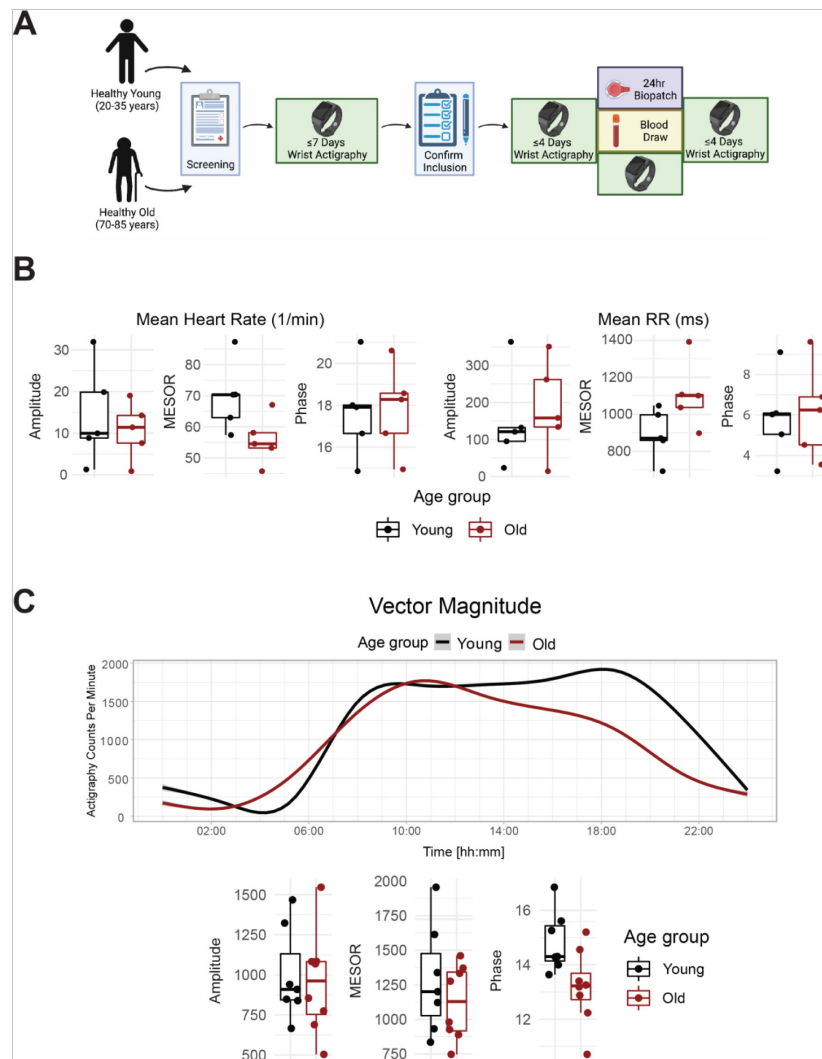


Fig 1.

Healthy elderly individuals tend to have lower heart rate, increased heart rate variability, and phase advanced activity patterns relative to young subjects.

Experimental protocol for enrolment and monitoring of human subjects (A, made with BioRender.com). As assessed by Zyphyr BioPatch, electrocardiogram (EKG) measurements suggest that MESOR of average heart rate decreases with age ($p=0.056$) and that MESOR of heart rate variability increases with age ($p=0.056$). $N=5$ per group. MESOR and amplitude were tested using two-sided Wilcoxon rank sum exact test, while phase was tested by Kuiper's two-sample test (B). Average activity counts across three axes (vector magnitude), as recorded by the actigraph device plotted throughout the day (top) and analyzed for circadian rhythm (bottom) (C). While the amplitudes of activity did not differ between the age groups, the older individuals trended towards an early phase ($p\sim 0.055$, Kuiper's two-sample test) compared to young individuals. $N=7$ for young and $N=8$ for old. Lines in the top panel of C are smoothed means (fit with penalized cubic regression splines) for data from each age group. Dots in the bottom panel of C are subject-level cosinor parameter estimates derived from cosinor fits to the actigraphy data. Boxplot midlines correspond to median values, while the lower and upper hinges correspond to the first and their quartiles, respectively. Boxplot whiskers extend to the smallest/largest points within $1.5 \times \text{IQR}$ (Inter Quartile Range) of the lower/upper hinge.

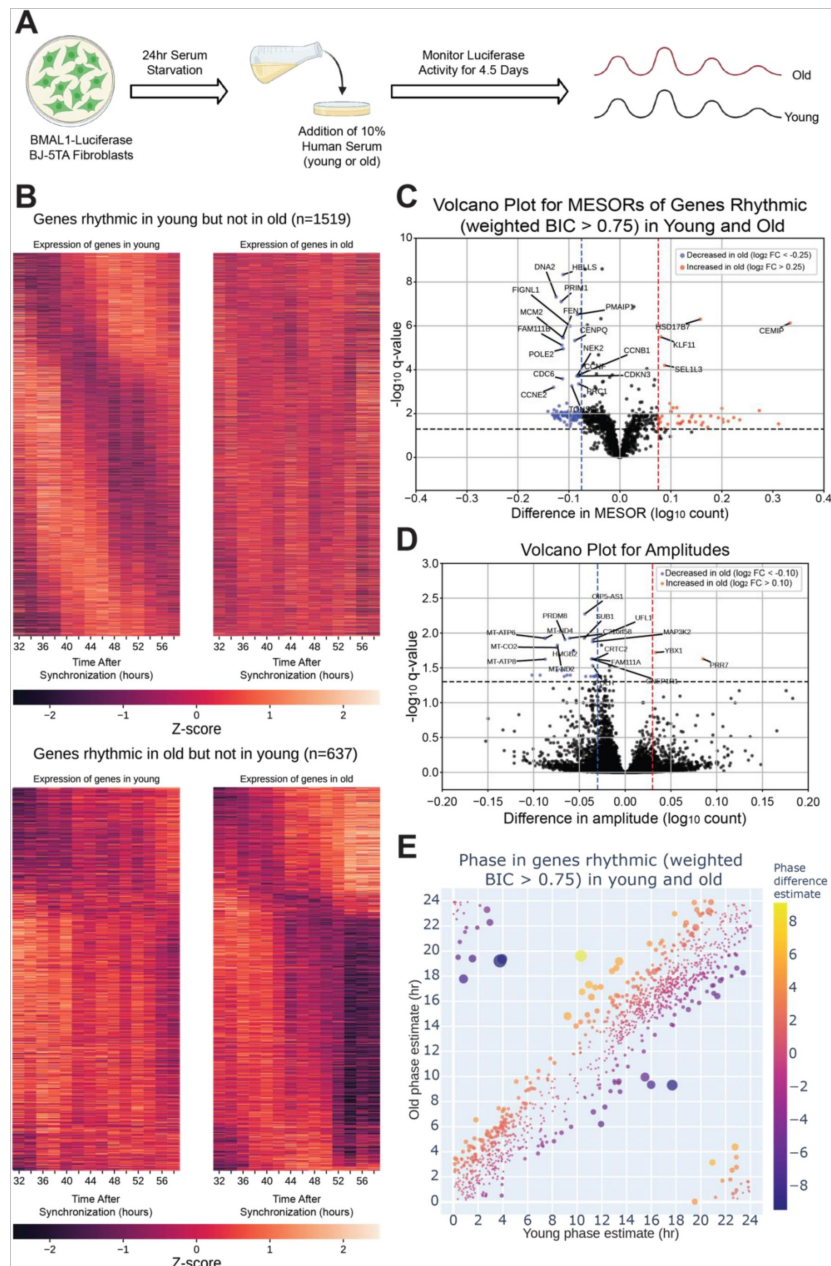


Fig 2.

The circadian transcriptome is differentially affected by entrainment with sera from young and old human subjects.

A visual representation of the serum starvation-serum addition protocol to synchronize the *BMAL1-luciferase* BJ-5TA fibroblasts (A, made with BioRender.com). When comparing the circadian transcriptome entrained by either young or old sera (n=4 sera per group), 1519 genes lost rhythmicity with age (B, top) while only 637 genes gained rhythmicity with age (B, bottom). Weighted BIC criterion with a threshold of 0.75 was used to assess rhythmicity (B). The number of genes rhythmic in young and old (according to weighted BIC > 0.75 criterion) that show a detectable change in MESOR ($q < 0.05$ criterion for MESOR difference) is 568, with MESOR increasing in 163 and decreasing in 405 genes. Out of these, 40 genes with increased MESOR (red) and 98 genes with decreased MESOR (blue) also satisfy the condition $|\log_2 FC| > 0.25$ (C). We were only able to detect change in amplitude for a small number of genes, 39 genes had decreased amplitude in old and 2 had increased amplitude using CircaCompare. Only 30 genes with decreased amplitude also satisfy the condition $|\log_2 FC| > 0.1$ (blue) (D). For phase, using a test provided by CircaCompare, under $q < 0.05$ cutoff for age-related phase differences in genes rhythmic in young and old (with BIC>0.75 cycling criteria), we detected 20 genes with advanced phase, and 34 with delayed phase (E).

these analyses low p-values can be driven by large sample-size such as in Fig S5 where distributions of MESORs and amplitudes across genes are assessed such that each gene represents a sample.

As mentioned, we found several examples of transcripts that cycle with young serum synchronization and lose rhythmicity (ex. *TCF4*), decrease in amplitude (ex. *HMGB2*), phase shift (ex. *TEF*), or change MESOR (ex. *HSD17B7*) with aged serum (**Fig. S6**). Search Tool for the Retrieval of Interacting Genes (STRING)¹⁴ and Ingenuity Pathway Analysis (IPA)¹⁵ were used for functional genomics. Both approaches indicate a maintenance of cycling of cell cycle genes with young and old serum synchronization, and a loss of rhythmicity of genes associated with oxidative phosphorylation in the aged serum (**Fig 3A** and S7). STRING analysis revealed that the dominant pathways associated with genes rhythmic in both young and old conditions, cell cycle and DNA replication, demonstrate a decrease in MESOR with old serum. For instance, checkpoint control and chromosomal replication pathway associated genes were expressed cyclically in both young and old conditions; however, several chromosomal replication pathway genes exhibit decreased MESORs in the aged sera (Fig S7 D,E, **Table S3**). MESORs of steroid biosynthesis genes, particularly those relating to cholesterol biosynthesis, were increased in the old sera condition (**Fig 3A**, **Table S4**).

Loss of rhythms in oxidative phosphorylation suggests mitochondrial dysfunction with age. By STRING analysis, 24 out of the 26 genes associated with oxidative phosphorylation overlap with the Alzheimer's Disease KEGG pathway highlighting the disease relevance of this class of genes that loses cycling in older individuals. Given that Alzheimer's pathology is closely associated with the accumulation of oxidative stress¹⁶, it is possible that loss of cycling contributes to oxidative damage. An additional 31 genes in the Alzheimer's Disease KEGG pathway lose rhythmicity with aged serum; these include amyloid precursor protein (*APP*) and apolipoprotein E (*APOE*). *APP* is by definition the precursor of amyloid beta, which accumulates in AD¹⁷. While *APOE* plays an important role in lipid transport, specific variants of this gene are strongly associated with AD risk as *APOE* interacts with amyloid beta in amyloid plaques, a hallmark of the disease¹⁸.

To determine whether aged serum modifies the activity of specific transcription factors, we performed an epigenetic Landscape In Silico deletion analysis (LISA), a computational tool designed to predict changes in transcription factor activity based on gene expression. Using a comprehensive database of known transcription factor binding profiles of fibroblasts, we identified altered gene expression corresponding to 59 total transcription factors ($q < 0.05$) in our young-versus-old cycling dataset. Potential changes in the activity of these transcription factors are associated with the following five categories of cycling phenotypes: decreased MESOR, increased MESOR, phase delay, gain of rhythmicity, and loss of rhythmicity, as were defined above (**Fig 3B**). 26 transcription factors were represented in all groups, and included those implicated in diseases such as cancer (*MYC*, *TP53*, *YAP1*, *RBL2*, *E2F7*, *BRD4*, *MXI1*), inflammation (*CEBPB*, *BHLHE40*), oxidative stress (*NFE2L2*), and neurological conditions (*NEUROG2*, *SUMO2*).

Lastly, several clock genes showed differences in expression with the aged serum, most notably genes in the Circadian Rhythm KEGG pathway (**Fig 4**). In particular, expression of *CRY1*, *CRY2*, *NR1D1*, *NR1D2*, *PER3*, and *TEF* was significantly phase delayed after synchronization with old serum compared to young (**Fig 4**). Importantly, the RNA-seq did not reveal a difference in the phase or amplitude of *BMAL1* expression with age, supporting the validity of our *BMAL1-luciferase* findings, although the MESOR significantly increased with age.

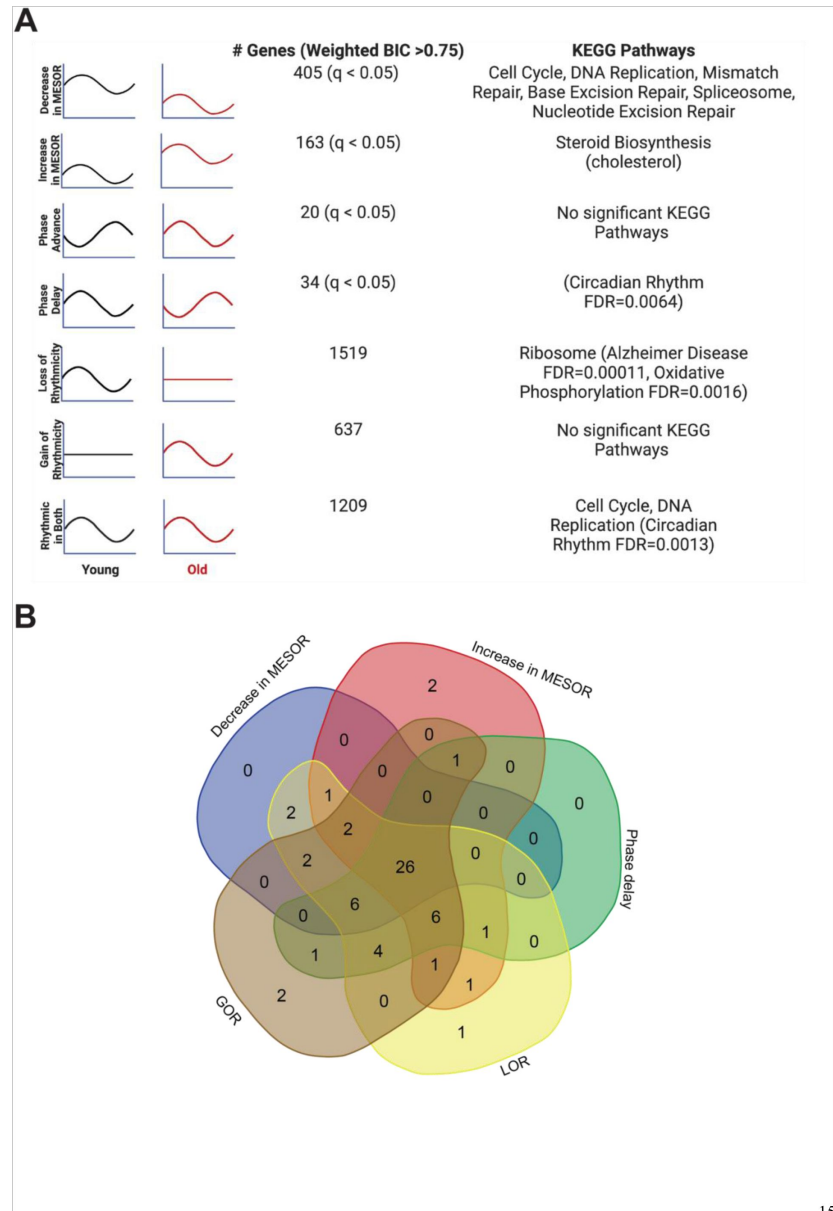


Fig 3.

Each type of circadian change is associated with different KEGG pathways by STRING analysis, but a similar set of transcription factors identified by LISA

Entrainment of the fibroblasts in culture with old serum significantly altered the circadian transcriptome despite use of the same cells in both young and old conditions, suggesting the aged serum itself affects the regulation of specific pathways in the cell. Significant KEGG Pathways (FDR < 1×10^{-4} unless specified) are indicated next to each category of genes. Age related pathways such as Alzheimer's Disease/oxidative phosphorylation are associated with a loss of rhythms in the old condition. Cell cycle and DNA replication pathways remain rhythmic in the old serum condition, but cycle with a decreased MESOR. Rhythmic genes were determined using CircaCompare. Rhythmicity was determined using the weighted BIC > 0.75 criterion while p-values for difference in MESOR and phase were determined using CircaCompare (A). LinC similarity analysis (LISA), based on known transcription factor binding in fibroblasts and RNA expression, suggests that 59 total transcription factors (q < 0.05) show significant changes in activity in conjunction with the following cycling phenotypes: decreased in MESOR, increased in MESOR, phase delay, gain of rhythmicity, and loss of rhythmicity as were defined above (B).

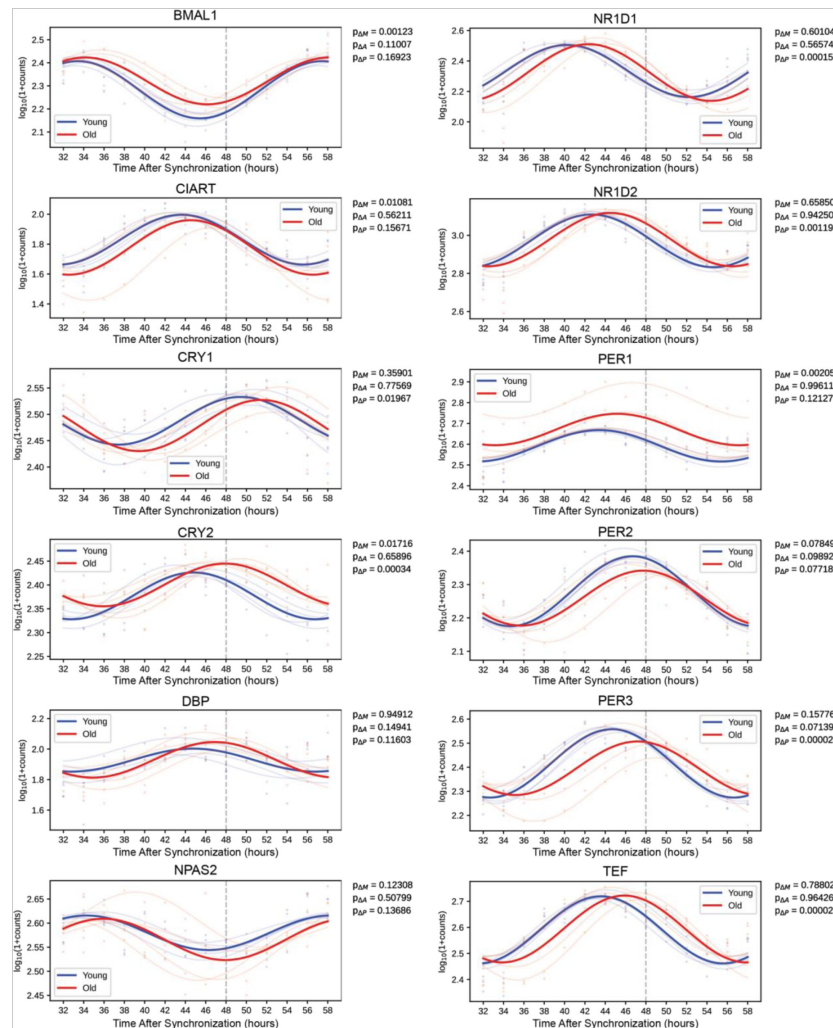


Fig 4.

Synchronizing with old serum phase delays the expression profile of several core clock genes

Traces of molecular clock mRNA transcripts, the average curve (bold) of results of individual sera (faded). P-values for the difference in MESOR ($P_{\Delta M}$), amplitude ($P_{\Delta A}$), and phase ($P_{\Delta P}$) are shown for the comparison of young and old conditions. Several clock genes are significantly phase delayed (*CRY1*, *CRY2*, *NR1D1*, *NR1D2*, *PER3*, *TEF*) in response to synchronization with old serum. While *BMAL1* is not phase delayed, the MESOR significantly increases with age. N=4 subjects per timepoint for both young and old groups.

Discussion

Studies of the aged SCN have revealed persistent cycling of clock gene expression but a breakdown of output signals^{4,5}. However, whether age-related changes in systemic signaling impact tissue/organ clocks has yet to be elucidated. We demonstrate here that while the core clock continues to cycle in cultured fibroblasts synchronized with serum from old volunteers, the circadian transcriptome is different from that seen in cells treated with serum from young individuals. Through this analysis of the role of serum in age-induced changes in circadian rhythms, we suggest a potential mechanism for observations where specific genes showed a loss, gain, or maintenance of rhythmicity with age^{19–23}. This phenomenon, known as circadian reprogramming, may illuminate which pathways are affected by or become more impactful for changes in cellular integrity through aging²⁴.

In order to study the effect of circulating factors on age-related changes of the circadian transcriptome, we utilized the well-established serum starvation-serum addition protocol⁷ to synchronize cells in culture. Cultured fibroblasts from old and young subjects have robust clocks and respond similarly to synchronization with dexamethasone⁹; however, when synchronized with dexamethasone in a media containing old serum, they reportedly exhibited shortened circadian periods⁹. This effect was reversed by heat inactivating old serum. We did not observe a significant difference in period length between young and old serum synchronization in our *BMAL1-luciferase* experiment, perhaps because dexamethasone was not utilized. Our use of serum to synchronize allowed us to more closely simulate *in vivo* conditions where blood is an important mode by which central clocks can entrain peripheral clocks to maintain synchrony with the day:night cycle. Previously, serum factors were shown to activate the immediate early transcription factor, SRF, in a diurnal manner to confer time of day signals to both cells in culture and to mouse liver⁷. We show here that effects of age are also mediated by serum.

We find that while the number of rhythmic transcripts (weighted BIC>0.75) in the young serum condition (~18%) is higher than the old serum condition (~12%), both conditions demonstrate a much larger number of cycling transcripts in these cultured fibroblasts than in other cell culture studies^{25–28}. In mammals, up to 20% of transcripts can cycle in a given tissue^{29,30}, and the low number of cycling transcripts in culture has been cited as a major limitation of the culture model²⁷. While our analysis used a normal human fibroblast cell line (BJ-5TA) we had higher statistical power, given that we had four different biological replicates at every 2-hour timepoint. However, a major contributing factor to the high rhythmicity might be the use of human serum as the synchronization signal. In this way, our model of mimicking signaling to peripheral tissues by using serum directly from humans may more accurately recapitulate the human condition. In this study we focused only on the effects of serum synchronization on fibroblasts, but it is likely that factors circulating in serum act on several tissues, and so their effects are relatively broad. However, we acknowledge that in order to support this claim future studies should investigate other peripheral tissue cell types. Additionally, in the future we intend to analyze the serum using a combination of fractionation and either proteomics or metabolomics to identify relevant factors for the regulation of peripheral rhythms.

Interestingly, many of the genes in the circadian transcriptome that exhibit age-related changes are independently implicated in aging. For instance, some of the genes that lose rhythmicity in the aged condition are involved in oxidative phosphorylation and mitochondrial function, both of which decrease with age³¹. Previous work in the field demonstrates that synchronization of the circadian clock in culture results in cycling of mitochondrial respiratory activity^{32,33} further underscoring the different effects of old serum, which does not support oscillations of oxidative phosphorylation associated transcripts. Age-dependent decrease in oxidative phosphorylation and increase in mitochondrial dysfunction³¹ is seen also in aged fibroblasts³⁴ and contributes to

age-related diseases³⁵. We suggest that the age-related inefficiency of oxidative phosphorylation is conferred by serum signals to the cells such that oxidative phosphorylation cycles are mitigated. On the other hand, loss of cycling could contribute to impairments in mitochondrial function with age.

Both pathway analyses utilized here identified increased MESORs of steroid biosynthesis components in the aged sera condition. In particular, the genes identified in this pathway are associated with cholesterol biosynthesis. Since elderly individuals typically have lower levels of cholesterol biosynthesis and higher levels of circulating cholesterol³⁶, we were surprised to see increased expression of biosynthesis transcripts. Perhaps the cholesterol synthesis related RNA levels are high as a response to low levels of cholesterol synthesis proteins within the cell. Previous studies in cultured hepatocytes demonstrated that increased reactive oxygen species resulted in higher levels of transcripts associated with cholesterol biosynthesis³⁷. Given that deficits in oxidative phosphorylation are already implicated in these findings, it is possible that oxidative stress plays a role in the increase of cholesterol biosynthesis transcripts.

Although our findings are largely supported by the aging literature, the relatively small sample size of our study necessitates follow up studies to control for individual differences between subjects. We observed variations in luminescence and transcript traces across individuals and while we did not see changes that could account for the overall significant differences in transcript cycling between young and old subjects, we cannot exclude the possibility that factors other than aging contribute to these data.

Together, these findings indicate that at least some of the age-related changes in the cultured fibroblast circadian transcriptome are derived from signals circulating in the serum and not the age of the cells. This has profound implications for understanding and treating circadian disruption with age, and could also be relevant for other age-related pathology, given established links between circadian disruption and diseases of aging. Notably, many of the genes whose cycling is affected by old serum contribute to age-associated disorders.

Materials and methods

Clinical research study

This clinical research study enrolled apparent healthy participants from the volunteer pool maintained by the Institute for Translational Medicine and Therapeutics (ITMAT), University of Pennsylvania. The Institutional Review Board of the University of Pennsylvania (Federal wide Assurance FWA00004028; IRB Registration: IORG0000029) approved the clinical study protocol (Penn IRB#832866). The study was registered on ClinicalTrials.gov with identifier NCT04086589. After obtaining informed consent from all volunteers, study assessments were conducted in the Center for Human Phenomic Science (Penn CHPS#3002) in accordance with relevant GCP guidelines and regulations. We originally intended on recruiting n=20 per group; however, patient recruitment was halted due to the COVID-19 pandemic. Participants met criteria for inclusion (in general good health, either 70-85 years of age for the elderly cohort or 20-35 years of age for the young cohort, and a wrist-actigraphy-based average TST (total sleep time) ≥ 6 hours per night occurring between 22:00 – 08:00). Participants were excluded due to pregnancy or nursing, shift work (defined as recurring work between 22:00-05:00), a history of clinically significant obstructive sleep apnea, transmeridian travel across ≥ 2 time zones in the two weeks prior to study assessments and one week after, > 2 drinks of alcohol per day, and use of illicit drugs. The subject-specific midsleep preferences were quantified using the Munich ChronoType Questionnaire (MCTQ)³⁸. Blood collections were done from the median cubital vein via venipuncture using a

22 G butterfly needle (BD, Franklin, Lakes, NJ, USA). All blood draws occurred at 14:00, the same time as Pagani et al.⁹. One subject returned for a repeat clinical assessment including biosampling to provide additional sample.

Acquisition of accelerometry data streams

Participants wore a triaxial actigraph device (wGT3X-BT, ActiGraph, Pensacola, FL) on the non-dominant wrist. The devices were initialized using the following parameters: start date and time were synchronized with atomic server time without pre-defined stop date/time, at 60 Hz sampling rate for the three accelerometer axes, enabled for delay modus, steps, lux, inclinometer, and sleep while active. Raw data were downloaded from the device in AGD and GT3X file format in one second epochs using ActiLife software (version 6, ActiGraph, Pensacola, FL) and submitted for further analyses. For visualization and cosinor analysis, actigraphy data were aggregated into 1-minute intervals by summing ActiGraph counts across each minute. Cosinor analyses of the data were adapted from the single component cosinor analysis reviewed by Cornelissen³⁹, as well as the cosine fit described by Refinetti et al.⁴⁰. Briefly, the measurement times for the actigraphy data were recalculated as hours since midnight on the first day of measurement, within each participant's data. For each actigraphy variable and each participant, the `lm()` function in R (v4.2.0) was used to perform a cosinor fit with a fixed 24-hour period. The two cosinor coefficients from these fits were used to calculate participant-level amplitudes and circadian phases, while the intercepts provided MESOR estimates. The two-sided Wilcoxon rank sum exact test, as implemented by R's `wilcox.test()` function, was used to test for significant differences in amplitude and MESOR between the age groups. A two-sample Kuiper's test, as implemented by the `kuiper_test(nboots = 10000)` function from the `twosamples` R package (v2.0.0), was used to test for significant differences in circadian phase between the age groups.

Acquisition of EKG data

The Zephyr BioPatch devices (Zephyr Technology, Annapolis, MD) were deployed as previously established⁴¹. All subjects included in this analysis wore the BioPatch for at least 24hrs. EKG recordings were analyzed by Kubios HRV Premium (ver. 3.5.0, Kubios Team, Kuopio, Finland) to report time-of-day-specific measures of cardiovascular function consisting of heart rate, RR intervals, sympathetic and parasympathetic nervous activity (SNS and PNS). Preprocessing in Kubios was set to automatic beat correction to remove artifacts and Smoothn priors for detrending. Cosinor fits and tests for differences in circadian parameters between age groups were performed as described for the actigraphy data. The circadian parameters MESOR and amplitude were tested for significant differences between the age groups by Wilcoxon rank sum exact test (two-sided), while phase was tested by Kuiper's two-sample test to account for the circular measurement.

Cortisol Measurements

Cortisol was measured in human serum by coated tube RIA (MP Bio, Solon OH) in duplicate. Tubes were counted on Perkin Elmer gamma counter and data reduced by STATLia software.

Generation of stable BJ-5TA cell

line expressing *BMAL1-dLuc-GFP*¹⁰

Virus was generated and cells were infected as we've previously described⁴². Briefly, LentiX 293T cells (Clontech) were transfected with Lipofectamine 3000 PLUS (Life Tech) using manufacturer instructions. The transfection included 18ug of DNA per reaction and the plasmid (*BMAL1-dLuc-GFP*) to packaging vector (DVPR, Addgene) to envelope (VSV-G, Addgene) ratio was 10:1:0.5. Media was changed 24hrs post-transfection after the cells were checked using a fluorescence scope to make sure cells were >50% GFP positive. Supernatant with the virus was collected at 48hrs and 72hrs post transfection and spun down at 3000 RPM for five minutes (to

eliminate any cells/debris). BJ-5TA cells were infected with fresh virus upon virus collection. Polybrene (Sigma-Aldrich, 10 mg/mL) was also added to aid with infection. Transduced BJ-5TA cells (>2000 cells) were sorted (FACSMelody, BD Biosciences) for high GFP expression. Once the cell line was established, blasticidin was added to the culture at 2ug/ml. Due to fragility of the cell line in the presence of antibiotic, BJ-5TA *Bmal1-luciferase* cells with differing expression levels of GFP were sorted. The line with highest stable luminescence oscillation was used for all experiments reported here.

Serum entrainment and bioluminescent recording

BJ-5TA *BMAL1-dLuc-GFP* cells in 24-well plates (~confluent) were washed (2x) with DPBS and given serum free media for 24hrs. After the starvation, cells were given media with 10% human serum and 200uM beetle luciferin potassium salt. Each well of cells was given media with the serum from a single patient. The cells were continuously monitored by LumiCycle luminometer (Actimetrics) for 4.5 days from the point of serum addition. LumiCycle raw data was exported using LumiCycle software (Actimetrics). The data were analyzed by BioDare2 (biodare2.ed.ac.uk⁴³) using FFT NLLS with baseline detrending. Any replicates that were not cycling or had a period outside of the 20-28hr range was excluded from analysis. Both the serum free media and serum added media used the recipe from¹⁰ at pH 7.4; however, the serum free media had no serum and the serum media had 10% human serum instead of FBS.

Sample preparation, RNA extraction, and RNA sequencing

BJ-5TA *BMAL1-dLuc-GFP* cells in 24-well plates (~confluent) were washed (2x) with DPBS and given serum free media (DMEM with Penn Strep) for 24hrs. After 24hrs cells were given media with human serum (DMEM, Penn Strep, 10% human serum). Serum starvation was staggered every 12 hours over two days to allow for samples to be collected on the same day. Upon sample collection wells were place on ice and rinsed with cold DPBS and then put in cold RLT buffer with 2-Mercaptoethanol (10μL/mL). Samples were frozen at -80 overnight and sent to Admera for RNA extraction and sequencing. Total RNA was extracted with RNeasy mini kit (Qiagen). Isolated RNA sample quality was assessed by High Sensitivity RNA Tapestation (Agilent Technologies Inc., California, USA) and quantified by Qubit 2.0 RNA HS assay (ThermoFisher, Massachusetts, USA). Paramagnetic beads coupled with oligo d(T)25 are combined with total RNA to isolate poly(A)+ transcripts based on NEBNext® Poly(A) mRNA Magnetic Isolation Module manual (New England BioLabs Inc., Massachusetts, USA). Prior to first strand synthesis, samples are randomly primed (5' d(N6) 3' [N=A,C,G,T]) and fragmented based on manufacturer's recommendations. The first strand is synthesized with the Protoscript II Reverse Transcriptase with a longer extension period, approximately 40 minutes at 42°C. All remaining steps for library construction were used according to the NEBNext® UltraTM II Non-Directional RNA Library Prep Kit for Illumina® (New England BioLabs Inc., Massachusetts, USA). Final libraries quantity was assessed by Qubit 2.0 (ThermoFisher, Massachusetts, USA) and quality was assessed by TapeStation D1000 ScreenTape (Agilent Technologies Inc., California, USA). Final library size was about 430bp with an insert size of about 300bp. Illumina® 8-nt dual-indices were used. Equimolar pooling of libraries was performed based on QC values and sequenced on an Illumina® NovaSeq S4 Illumina, California, USA) with a read length configuration of 150 PE for 40 M PE reads per sample (20 M in each direction).

RNA-seq and statistical analysis

Raw RNA-seq reads were aligned to the GRCh38 build of the human genome by STAR version 2.7.10a⁴⁴. The dataset contained an average of 19,265,220 paired-end non-stranded 150 bp reads mapping uniquely to genes, per sample. Data were normalized and quantified at both gene and exon-intron level, using a downsampling strategy implemented in PORT (Pipeline Of RNA-seq

Transformations, available at <https://github.com/itmat/Normalization>, version 0.8.5f-beta_hotfix1. Both STAR and PORT were provided with gene models from release 106 of the Ensembl annotation⁴⁵.

MESOR, amplitude, and phase estimates, as well as p-values for the difference in MESOR, amplitude, and phase, were calculated with CircaCompare¹³, version 0.1.1. Only rhythmic genes were taken into consideration in pathway and other analyses involving MESORs and phases. The criterion for rhythmicity was either based on (BH adjusted) p-values reported by CircaCompare, or weighted BIC values, obtained by an approach similar to dryR⁴⁶. In the latter approach we fitted four models of rhythmicity, one modeling gene expression that is rhythmic in both cells treated with young or old sera, another modeling gene expression rhythmic in neither cells treated with young nor old sera, and two modeling gene expression rhythmic in either young or old sera treated cells respectively. The BIC values were calculated for the four models and were weighted to obtain numbers between 0 and 1. All methods were provided with $\log_{10}(1 + \text{PORT normalized count})$ values and were run on R, version 4.1.2, accessed through Python, version 3.9.9, via rpy2, an interface to R running embedded in a Python process, <https://rpy2.github.io/>, version 3.4.5. Additionally, we used Nitecap⁴⁷ to visualize and explore circadian profiles of gene expression.

STRING pathway analysis

We performed STRING¹⁴ pathway analyses using STRING API version 11.5. Enrichment analyses were performed on the sets of genes rhythmic in both groups (weighted BIC > 0.75) with observed decrease in MESOR, increase in MESOR, advance in phase, and delay in phase according to CircaCompare $q < 0.05$ criterion. We also performed analyses on the sets consisting of genes rhythmic only in young group (weighted BIC > 0.75), only in old group (weighted BIC > 0.75), and the set of genes rhythmic in both groups (weighted BIC > 0.75). Finally, two additional analyses were performed, on the sets of genes with observed decrease and increase of amplitude (CircaCompare $q < 0.05$).

Ingenuity Pathway Analysis

QIAGEN IPA¹⁵ was used to identify pathways enriched in various subsets of cycling genes. The following subsets of genes were identified using a combination of CircaCompare stats and BIC cutoffs: (1) Decreased MESOR in old sera (929 genes) – CircaCompare rhythmic $q < 0.05$ in old, CircaCompare rhythmic $q < 0.05$ in young, CircaCompare MESOR difference $q < 0.05$, MESOR difference (old – young) < 0. (2) Increased MESOR in old sera (515 genes) – same selection criteria as ‘Decreased MESOR in old sera,’ except MESOR difference (old – young) > 0. (3) Phase advance in old sera (148 genes) - CircaCompare rhythmic $q < 0.05$ in old, CircaCompare rhythmic $q < 0.05$ in young, CircaCompare Phase difference $q < 0.05$, phase difference (old – young) < 0. (4) Phase delay in old sera (156 genes) – same selection criteria as ‘Phase advance in old sera,’ except phase difference (old – young) > 0. (5) Loss of rhythmicity in old sera (1519 genes) – weighted BIC > 0.75 for ‘Rhythmic in Young but not Old’ model. (6) Gain of rhythmicity in old sera (637 genes) – weighted BIC > 0.75 for ‘Rhythmic in Old but not Young’ model. (7) Rhythmic in old and young sera (1209 genes) – weighted BIC > 0.75 for ‘Rhythmic in both Old and Young’ model. Each of these gene subsets were processed separately with IPA’s core analysis, using default parameters.

For visualization via heatmap, we perform three rounds of normalization within each gene. First, we mean-normalize the read counts within each age group and serum treatment group (A, B, C, D). This is to account for baseline differences between sera collected from the different subjects. Second, we collapse replicates at each timepoint by calculating their means. Third, we calculate Z-Scores across all timepoints, within each age group. Note, these normalization procedures are to aid with visualization of the data and were not used as part of the statistical analyses.

Epigenetic Landscape in Silico deletion (LISA)

LISA analysis⁴⁸ was used to perform transcription factor binding analysis. LISA results were filtered by fibroblast. $q < 0.05$ cut off was used within each of the 5 groups. Venn diagram was generated with <https://bioinformatics.psb.ugent.be/webtools/Venn/>

Data and materials availability

Sequencing data are deposited in Gene Expression Omnibus (NCBI) under accession number GSE270290. All additional data files are available upon request.

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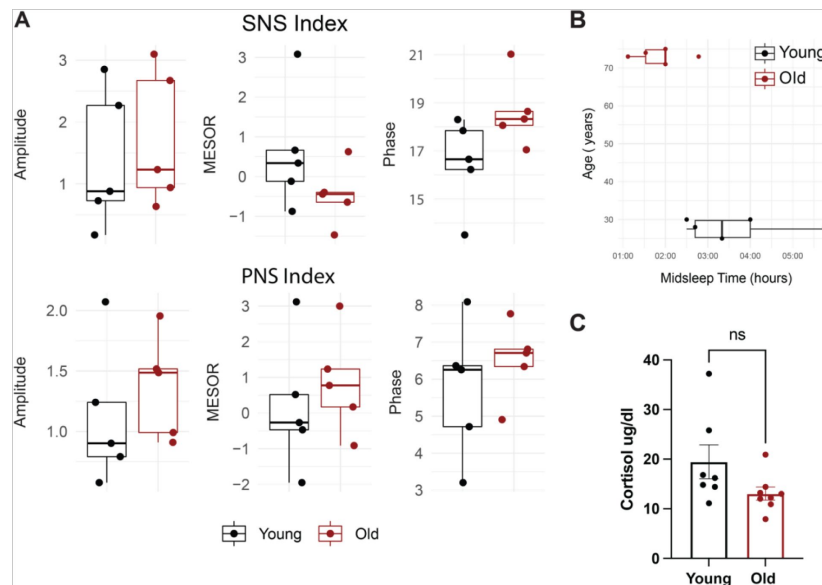


Fig S1.

There are differences in midsleep time, but no significant differences in sympathetic, parasympathetic nervous system indices, or cortisol levels between young and old individuals

Sympathetic nervous system (SNS, A, top) and parasympathetic nervous system (PNS, A, bottom) indices were measured using EKG. Amplitude, MESOR, and Phase were calculated from the resulting oscillations and the Median, quartiles, and SEMs are shown. Dots represent individuals. $P > 0.05$. $N = 5$ per group. MESOR and amplitude were tested using Wilcoxon rank sum exact test, while phase was tested by Kuiper's two-sample test. Boxplot midlines correspond to median values, while the lower and upper hinges correspond to the first and third quartiles, respectively. Boxplot whiskers extend to the smallest/largest points within $1.5 \times \text{IQR}$ (inter Quartile Range) of the lower/upper hinge (A). Boxplot distributions of midsleep times for subjects in the young (black) and old (red) cohorts. The midsleep time is advanced in the old subjects compared to young by Mann-Whitney U test (Wilcoxon rank sum test) $p = 0.036$. We calculated midsleep times for each subject from their responses to the Munich Chronotype Questionnaire (MCTQ). $N = 5$ per group. We were not able to calculate midsleep times for four subjects (two from the old cohort and two from the young cohort) because they used alarms to wake on their non-working days and one old subject because they exhibited a total sleep time less than 6 hours (B). Serum cortisol levels are not statistically different by unpaired t-test (C, $N = 7$ young, 8 old).

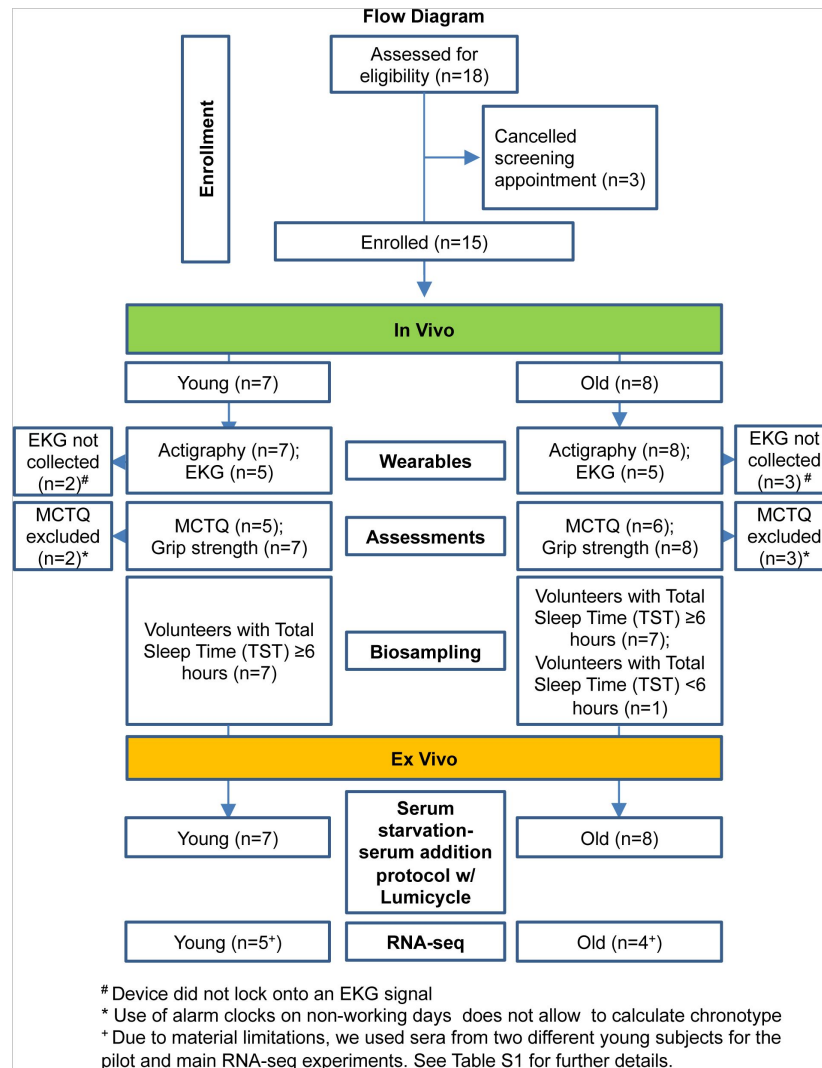


Figure S2.

Subject Inclusion Flow Chart

A flow chart of the number of subjects included in each analysis present in this study.

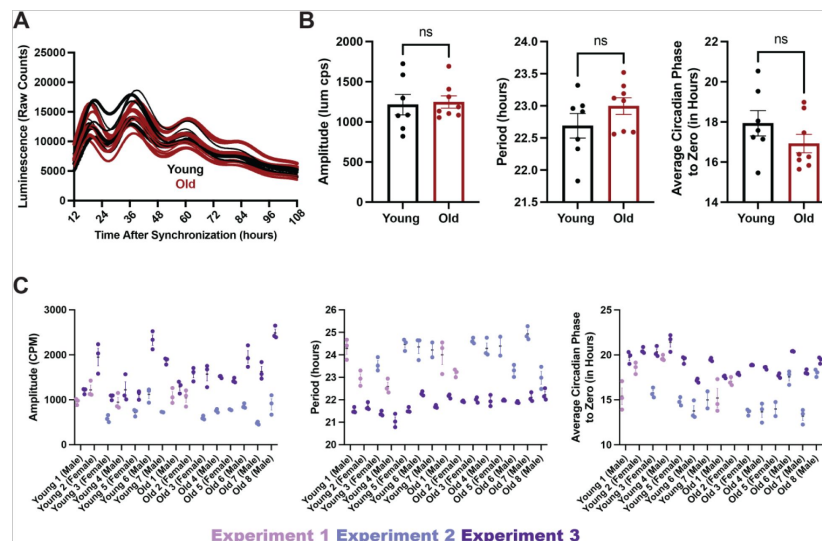


Fig S3

Young and old serum are equally effective at entraining cells in culture

(A,B) Cells synchronized with human serum from either young (N=7 subjects) or old (N=8 subjects) individuals did not show differences in amplitude (as measured in luminescence counts per second, bottom, left), period (bottom, middle), or phase (bottom, right). Line traces and data points represent average values for a specific patient's serum over 2 experiments with 2-3 replicates per experiment. A visual representation of the individual replicates averaged in B **(C)** Summary statistics are displayed as mean $\pm$ SEM. Means compared by unpaired t-test. Despite individual and run to run variability, cells synchronized with human serum from either young (N=7 subjects) or old (N=8 subjects) individuals showed similar amplitude (left), period (middle), and phase relative to synchronization time (right) of BMAL1-luciferase rhythms. Data points represent individual replicates within an experiment. Wells run in the same experiment are displayed in the same color. Each subject's sample was run in 2-3 replicates.

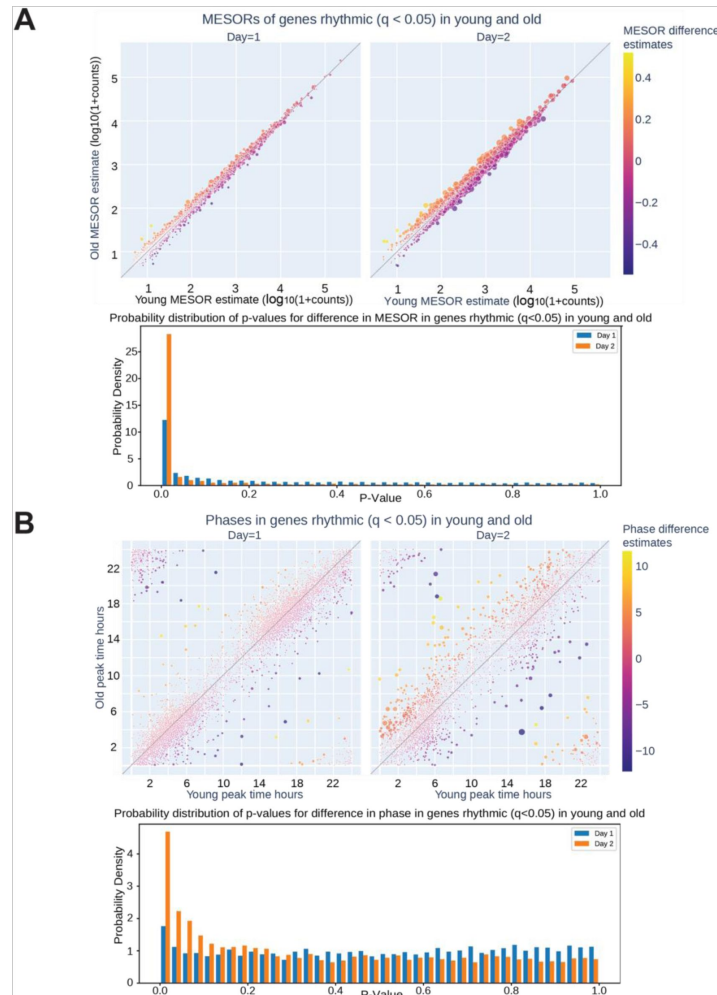


Fig S4.

The circadian transcriptomes of cells synchronized with young or old sera deviate significantly on day two.

On day two of serum entrainment, young and old transcriptional rhythms differentiated. CircaCompare analysis of RNA sequencing revealed that on Day 2 of serum entrainment analysis (36-58 hours after synchronization) the MESOR differences of cycling genes are larger between the cells entrained with young or old serum (**A**, top). Additionally, more genes were phase shifted in the old serum condition compared to young serum on Day 2 (**B**, top). The distribution of p-values shows an enrichment of low p-values on Day 2 for both MESOR and phase differences (**A**, bottom, **B**, bottom). The size of each circle in the scatter plots is proportional to $-\log_{10}q$, hence bigger circles correspond to smaller q-values for the difference between young and old for each metric.

Figure S5

There is a small effect of age on the amplitude and MESOR of cycling mRNA between young and old serum samples.

The difference (Top, left) between young and old mRNA amplitude values (bottom, left) are significantly different by Wilcoxon signed-rank test ($p = 3.73 \times 10^{-39}$). The difference (Top, right) between young and old mRNA MESOR values (bottom, right) are significantly different by Wilcoxon signed-rank test ($p = 6.94 \times 10^{-22}$). However, these small differences result in very low p-values due to the large number of genes inspected.

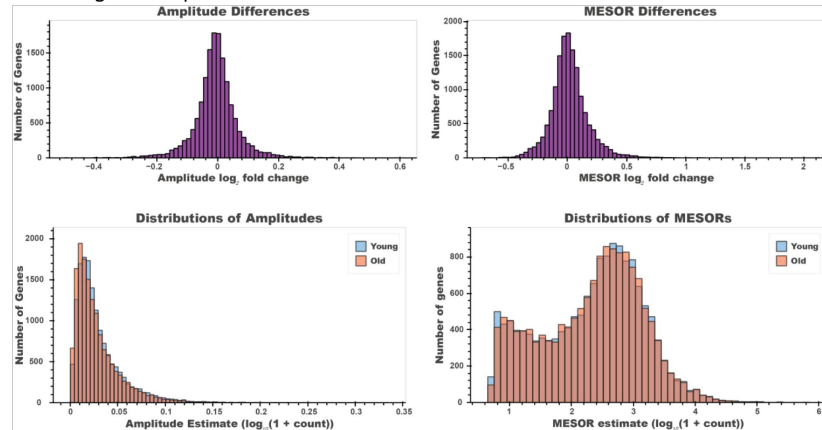
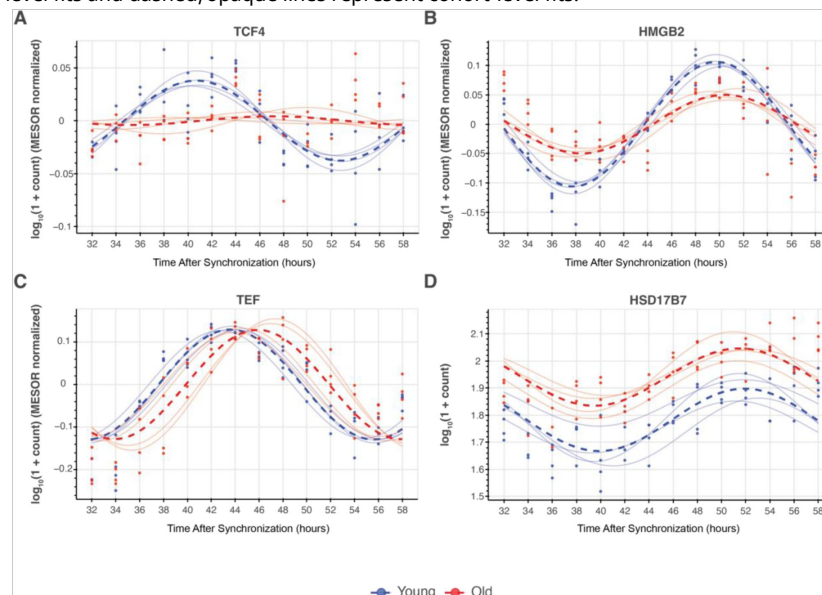


Figure S6

Sample traces of transcripts that cycle with young serum synchronization and are differentially affected by old serum synchronization.

Example traces of mRNA transcripts that lose cycling (A, TCF4), decrease amplitude (B, HMGB2), phase shift (C, TEF), and increase MESOR (D, HSD17B7). To ease the comparison of amplitudes and phases in traces depicted in (A-C), subject-level differences in MESORs were removed by subtracting the measured gene expression of each subject at each timepoint by the MESOR of the subject's gene expression oscillation pattern, making all subject-level traces oscillate around zero. Traces in (D) were not subjected to this transformation since (D) showcases differences in MESORs themselves. Transparent lines represent subject-level fits and dashed/opaque lines represent cohort-level fits.



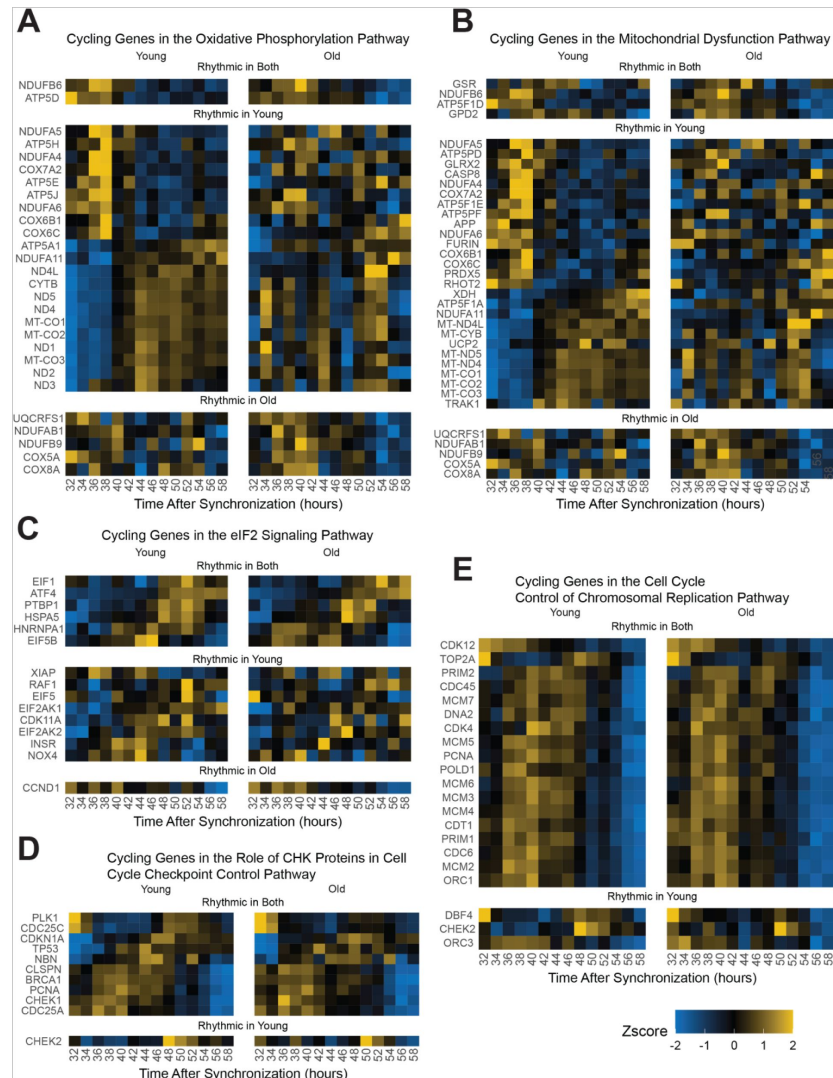


Fig S7

Oxidative phosphorylation genes and chromosomal replication genes are affected by entrainment with old serum

Additional pathway analysis employing IPA further emphasizes finding with STRING analysis. Genes identified by IPA analysis to be associated with oxidative phosphorylation/mitochondrial dysfunction and eIF2 genes lose rhythmicity with age (BIC >0.75) (**A, B, C**). Cell cycle checkpoint control (CHK proteins) (**D**) and chromosomal replication genes (**E**) maintain their cycling in the aged condition, which is consistent with the continued division of cells.

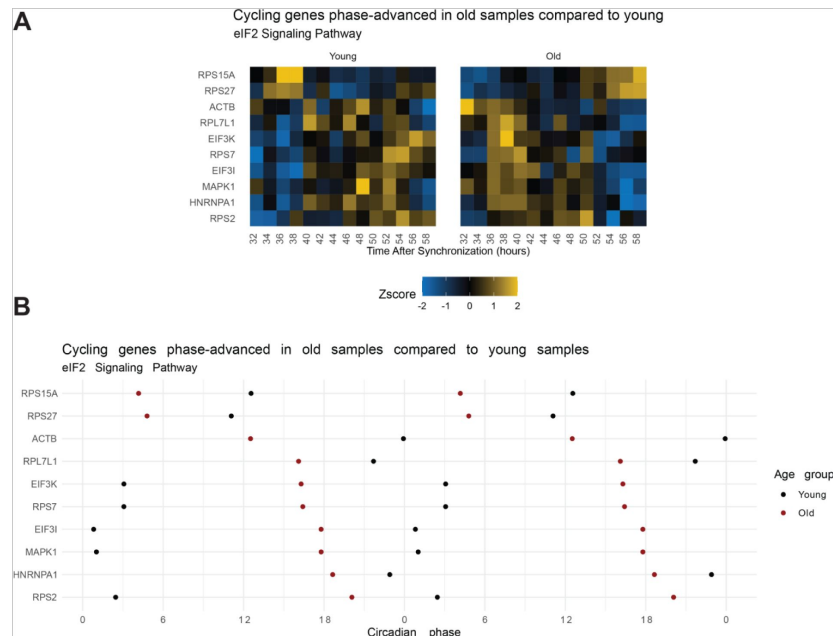


Fig S8

eIF2 signaling pathway genes that maintain cycling with age phase advance in the old serum condition

From the IPA analyses, the eIF2 signaling pathway was enriched ($BIC > 0.75$) in two datasets: 1) genes cycling in the young sera and not the old and 2) cycling genes phase-advanced in the old sera ($q < 0.05$). A significant number of genes that maintain rhythmicity in the old sera condition are phase advanced as visualized in the heat map (A) or plot of peak phase (B). Note, the data in the peak phase plot are double-plotted. This means the range of the x-axis is doubled and the datapoints between 0 and 24 are duplicated in the right half of the plot. Phase is a circular metric meaning a phase of 0 is equivalent to a phase of 24 ($1 \Rightarrow 25$, $2 \Rightarrow 26$, etc). Double-plotting is used to visualize data with repeated patterns or when data straddle the 24-0 boundary. As the eIF2 pathway is involved in protein translation, this suggests that the timing of protein translation is phase advanced in the older serum and/or less rhythmic.

Age Group	Sex	Age (y)	BMI (kg/m ²)	Total Sleep Time (TST) [h]	Actigraphy	MCTQ	Grip Strength	EKG	RNA-seq: Main	RNA-seq: Pilot (Fig S4)
20-35	Female	25	23.7	6.6	Yes	Yes	Yes	No	No	No
	Female	28	21.2	6.9	Yes	No	Yes	Yes	Yes	No
	Male	28	25	9.7	Yes	Yes	Yes	Yes	Yes	No
	Male	28	25.2	7.4	Yes	No	Yes	No	Yes	No
	Male	28	25.9	9.3	Yes	Yes	Yes	Yes	No	No
	Female	30	28.7	7.4	Yes	Yes	Yes	Yes	Yes	No
	Male	30	25.9	6.4	Yes	Yes	Yes	Yes	No	Yes
	Mean	28.1	25.1	7.7	—	—	—	—	—	—
Young	SD	1.7	2.3	1.3	—	—	—	—	—	—
70-85	Male	70	26.8	6.7	Yes	No	Yes	Yes	No	No
	Male	71	26.7	11.3	Yes	Yes	Yes	No	Yes	No
	Female	73	32.4	7.7	Yes	Yes	Yes	No	Yes	No
	Male	73	21.5	6.6	Yes	Yes	Yes	Yes	No	No
	Female	74	26.4	7.5	Yes	No	Yes	No	No	No
	Male	74	27.2	7	Yes	Yes	Yes	Yes	Yes	Yes
	Male	75	28.8	7.8	Yes	Yes	Yes	Yes	No	No
	Female	76	22.3	5.8	Yes	Yes	Yes	Yes	Yes	No
Old	Mean	73.3	26.5	7.6	—	—	—	—	—	—
	SD	2	3.4	1.7	—	—	—	—	—	—

Total sleep time (TST) was calculated from ActiLife 6 determined in-bed and out-of-bed times averaged across nights with available actigraphy data (≥ 7 nights). Old (N=8), Y-young (N=7). For the RNAseq we did a pilot experiment which had one young and one old (Fig S4). The young sample didn't have enough serum left to include that individual in the actual experiment (n=4 per group), but the old subject did. So, one old subject was used in both figures and the young were different.

Table S1.

Demographic information for subjects in the study.

ID	BioPatch EKG data collected?	Reason why BioPatch EKG data was not collected
Old 1	Yes	
Old 2	No	Device was deployed to participant. When device was returned, we found that data had not been collected during the observational period.
Old 3	No	Device was deployed to participant. When device was returned, we found that data had not been collected during the observational period.
Old 4	Yes	
Old 5	Yes	
Old 6	No	Device was deployed to participant. When device was returned, we found that data had not been collected during the observational period.
Old 7	Yes	
Old 8	Yes	
Young 1	Yes	
Young 2	No	Device was deployed to participant. When device was returned, we found that data had not been collected during the observational period.
Young 3	Yes	
Young 4	Yes	
Young 5	Yes	
Young 6	Yes	
Young 7	No	Device was deployed to participant. When device was returned, we found that data had not been collected during the observational period.

A log of which participants were included in our biopatch data based on equipment function.

Table S2.

BioPatch EKG Data Inclusion Log.

Table S3

Genes involved in the IPA cell cycle of chromosome replication pathway show decreased MESOR with age

Table of genes that are rhythmic in both conditions with a decreased MESOR with old serum treatment compared to young serum treatment by CircaCompare. All genes are involved in the cell cycle/DNA replication pathway.

Gene	Symbol	Old_mesor_estimate	Young_mesor_estimate
ENSG00000014138	POLA2	1.942220257	2.030676546
ENSG00000062822	POLD1	2.727523056	2.828026817
ENSG00000073111	MCM2	3.003984606	3.115331295
ENSG00000076003	MCM6	3.124116273	3.203138512
ENSG00000085840	ORC1	2.177440998	2.305154417
ENSG00000091651	ORC6	2.415062674	2.513336383
ENSG00000093009	CDC45	2.428834807	2.540049538
ENSG00000094804	CDC6	2.566892644	2.679401971
ENSG00000097046	CDC7	2.279003902	2.377425914
ENSG00000100297	MCM5	3.17577594	3.288560637
ENSG00000101868	POLA1	2.640718445	2.728951631
ENSG00000104738	MCM4	3.310670684	3.413600036
ENSG00000112118	MCM3	3.168296215	3.269321473
ENSG00000117748	RPA2	2.754645539	2.798914718
ENSG00000131747	TOP2A	3.722304752	3.798890616
ENSG00000132383	RPA1	3.238584015	3.287556646
ENSG00000132646	PCNA	3.232760966	3.321439626
ENSG00000135446	CDK4	3.143303063	3.172936739
ENSG00000138346	DNA2	2.113453067	2.238476228
ENSG00000146143	PRIM2	2.620994795	2.675363852
ENSG00000164815	ORC5	2.361538047	2.414675015
ENSG00000166508	MCM7	3.381394362	3.482444492
ENSG00000167258	CDK12	3.10241713	3.129181459
ENSG00000167513	CDT1	2.474002764	2.590319547
ENSG00000198056	PRIM1	2.116803844	2.231962095

mn

Table S4

Genes with increased MESOR in the IPA cholesterol biosynthesis pathway

Table of genes that are rhythmic in both conditions with a decreased MESOR with old serum treatment compared to young serum treatment by CircaCompare. All genes are involved in the cholesterol biosynthesis pathway.

Gene	Symbol	Old_mesor_estimate	Young_mesor_estimate
ENSG00000001630	CYP51A1	2.040377336	1.871207095
ENSG00000079459	FDFT1	3.330291993	3.19001773
ENSG00000104549	SQLE	3.116822033	2.950386727
ENSG00000110921	MVK	2.530172998	2.4121956
ENSG00000112972	HMGCS1	2.93111747	2.696086286
ENSG00000113161	HMGCR	3.011982868	2.833316048
ENSG00000116133	DHCR24	3.506294106	3.343492054
ENSG00000120437	ACAT2	2.71791125	2.58109012
ENSG00000132196	HSD17B7	1.940436315	1.782521965
ENSG00000147383	NSDHL	2.540302415	2.468852126
ENSG00000160285	LSS	3.486802994	3.411431601
ENSG00000160752	FDPS	3.143592562	3.081359846
ENSG00000167508	MVD	2.819963909	2.68386468
ENSG00000172893	DHCR7	2.990620862	2.797567922

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47. Brooks T. G., et al. (2022) **Nitecap: An Exploratory Circadian Analysis Web Application** *J Biol Rhythms* **37**:43–52
48. Qin Q., et al. (2020) **Lisa: inferring transcriptional regulators through integrative modeling of public chromatin accessibility and ChIP-seq data** *Genome Biol* **21**

Editors

Reviewing Editor

Kristin Tessmar-Raible

University of Vienna, Vienna, Austria

Senior Editor

Claude Desplan

New York University, New York, United States of America

Reviewer #1 (Public review):

Aging is associated with a number of physiologic changes including perturbed circadian rhythms. However, mechanisms by which rhythms are altered remain unknown. Here authors tested the hypothesis that age-dependent factors in the sera affect the core clock or outputs of the core clock in cultured fibroblasts. They find that both sera from young and old donors are equally potent at driving robust ~24h oscillations in gene expression, and report the surprising finding that the cyclic transcriptome after stimulation by young or old sera differs markedly. In particular, genes involved in the cell cycle and transcription/translation remain rhythmic in both conditions, while genes associated with oxidative phosphorylation and Alzheimer's Disease lose rhythmicity in the aged condition. Also, the expression of cycling genes associated with cholesterol biosynthesis increases in the cells entrained with old serum. Together, the findings suggest that age-dependent blood-borne factors, yet to be identified, affect circadian rhythms in the periphery. The most interesting aspect of the paper is that the data suggest that the same system (BJ-5TA), may significantly change its rhythmic transcriptome depending on how the cells are synchronized. While there is a succinct discussion point on this, it should be expanded and described whether there are parallels with previous works, as well as what would be possible mechanisms for such an effect.

Comments on revised version:

The authors have done a thorough revision of their manuscripts and provided convincing answers to all of my points. In particular, I applaud the authors for having added raw luminescence traces, and for providing Figure S5 on the amplitudes. Perhaps the authors could add a comment in the final text that the amplitudes are fairly low, $10^{0.1} = 1.25$ which means that the bulk of those genes has rhythms of at most 25%, which could reflect that the synchronization of the cells is partial.

<https://doi.org/10.7554/eLife.88322.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Aging is associated with a number of physiologic changes including perturbed circadian rhythms. However, mechanisms by which rhythms are altered remain unknown. Here authors tested the hypothesis that age-dependent factors in the sera affect the core clock or outputs of the core clock in cultured fibroblasts. They find that both sera from young and old donors are equally potent at driving robust ~24h oscillations in gene expression, and report the surprising finding that the cyclic transcriptome after stimulation by young

or old sera differs markedly. In particular, genes involved in the cell cycle and transcription/translation remain rhythmic in both conditions, while genes associated with oxidative phosphorylation and Alzheimer's Disease lose rhythmicity in the aged condition. Also, the expression of cycling genes associated with cholesterol biosynthesis increases in the cells entrained with old serum. Together, the findings suggest that age-dependent blood-borne factors, yet to be identified, affect circadian rhythms in the periphery. The most interesting aspect of the paper is that the data suggest that the same system (BJ-5TA), may significantly change its rhythmic transcriptome depending on how the cells are synchronized. While there is a succinct discussion point on this, it should be expanded and described whether there are parallels with previous works, as well as what would be possible mechanisms for such an effect.

We've expanded our discussion in the manuscript to discuss possible mechanisms and also how the genes/pathways implicated in our study relate to other aging literature.

Major points:

Fig 1 and Table S1. Serum composition and levels of relevant blood-borne factors probably change in function of time. At what time of the day were the serum samples from the old and young groups collected? This important information should be provided in the text and added to Table S1.

We made sure to highlight the collection time in the abstract of the manuscript "We collected blood from apparently healthy young (age 25-30) and old (age 70-76) individuals at 14:001 and used the serum to synchronize cultured fibroblasts." The time of blood draw is also in sections of the paper (Intro and Methods). Since Table S1 is demographic information, we did not think that the blood draw time fit best there, but hopefully it is now clear in the text.

Fig 2A. Luminescence traces: the manuscript would greatly benefit from inclusion of raw luminescence traces.

Raw luminescence traces have been added to Figure S3 (S3A).

Fig 2. Of the many genes that change their rhythms after stimulation with young and old sera, what are the typical fold changes? For example, it would be useful to show histograms for the two groups. Does one group tend to have transcript rhythms of higher or lower fold changes?

We've presented these data in Figure S5. There are a few significant differences, but largely the groups are similar in terms of fold change.

Fig. 2 Gene expression. Also here, the presentation would benefit from showing a few key examples for different types of responses.

Sample traces of genes that gain rhythmicity, lose rhythmicity, phase shift, and change MESOR are now illustrated in Figure S6.

What was the rationale to use these cells over the more common U2OS cells? Are there similarities between the rhythmic transcriptomes of the BJ-5TA cells and that of U2OS cells or other human cells? This could easily be assessed using published datasets.

The original rationale to use BJ-5TA fibroblast cells was that we were aiming to build upon an observation found in a previous study² which showed that circadian period changes with age in human fibroblasts. While our findings did not match theirs, we think an added benefit of

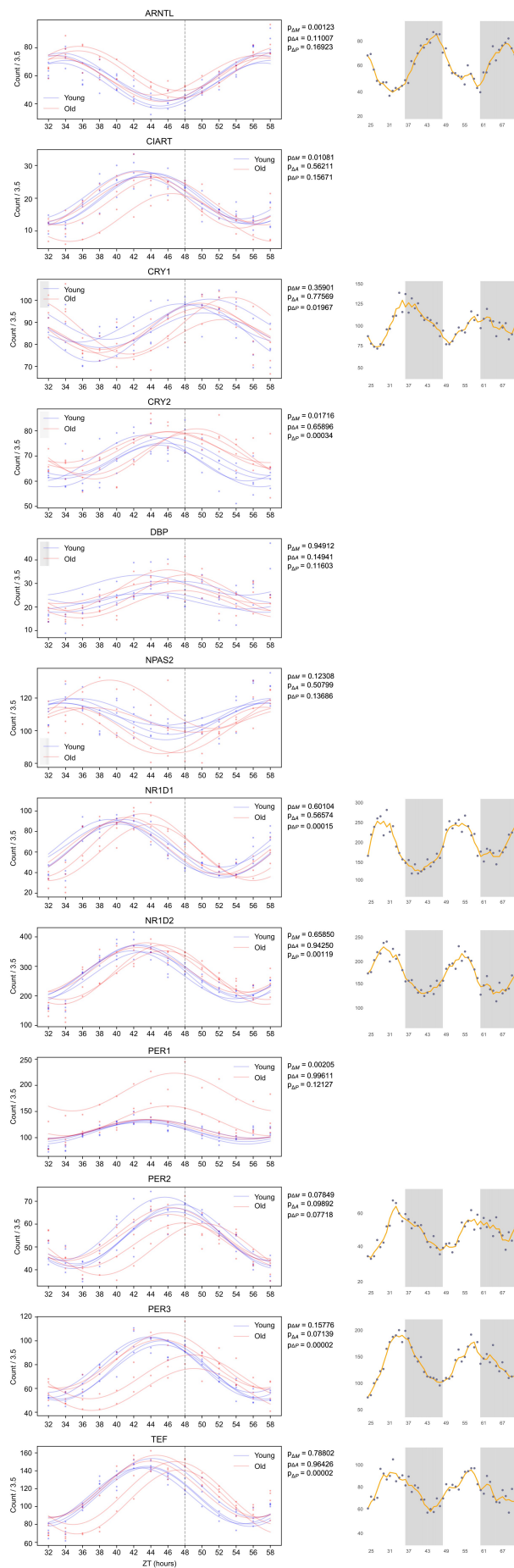
using the BJ-5TA line is that unlike U2OS cells, it is not a carcinoma derived cell line. We've added this point in lines 98-101.

Our study finds many more rhythmic transcripts compared to the previous studies examining U2OS cells. This can be attributed to several factors including differences in methods, including the use of human serum in our study, cell type differences, or decoupling of rhythms in some cancer cells. While a comparison of BJ-5TA cells and U2OS cells could be interesting, a proper comparison requires investigation of many data sets, since any pair of BJ-5TA and U2OS data sets will most likely differ in some detail of experimental design or data processing pipeline, which could contribute to observed differences in rhythmic transcripts.

That being said, we compared clock reference genes (see Author response image 1) between BJ-5TA and U2OS cells, comparing circadian profiles obtained from our data with those available on CircaDB. These circadian profiles exhibit many similarities and a few differences. The peak to trough ratios (amplitudes) are quite similar for ARNTL, NR1D1, NR1D2, PER2, PER3, and are about 25% lower for CRY1 and somewhat higher for TEF (about 15%) in our data. We find that the MESORS are generally similar with the exception of NR1D1 which is much lower and NR1D2 which is much higher in our data.

Author response image 1.

BJ-5TA and U2OS Cells Exhibit Similar Profiles of Circadian Gene Transcription. We compared the transcriptomic profiles of the BJ-5TA cells in young and old serum (left) to the U2OS transcriptomic data (right) available on CircaDB, a database containing profiles of several circadian reference genes in U2OS cells. This figure suggests that circadian profiles of these genes exhibit many similarities. We find that the peak to trough ratios (amplitudes) are similar for ARNTL, NR1D1, NR1D2, Per2, PER3, and that the MESORS are similar (with the exception of NR1D1 which is much lower and NR1D2 which is much higher in the BJ-5TA cells). We find that the amplitudes of CRY1 is ~25% lower and TEF is ~15% higher for the BJ5TA cells. The axis for plots on the left show counts divided by 3.5 in order to made MESORS of ARNTL similar to ease comparison.



For the rhythmic cell cycle genes, could this be the consequence of the serum which synchronizes also the cell cycle, or is it rather an effect of the circadian oscillator driving rhythms of cell cycle genes?

This is an interesting point. Given our previous data showing that the cell cycle gene cyclin D1 is regulated by clock transcription factors³, we believe the circadian oscillator drives, or at least contributes, to rhythms of cell cycle genes. However, the serum clearly makes a difference as we find that MESORs of cell cycle genes decrease with aged serum. This is consistent with the decreased proliferation previously observed in aged human tissue⁴.

While the reduction of rhythmicity in the old serum for oxidative phosphorylation transcripts is very interesting and fits with the general theme that metabolic function decreases with age, it is puzzling that the recipient cells are the same, but it is only the synchronization by the old and young serum that changes. Are the authors thus suggesting that decrease of metabolic rhythms is primarily a non cell-autonomous and systemic phenomenon? What would be a potential mechanism?

We are indeed suggesting this, although it is also possible that it is not cycling *per se*, but rather an overall inefficiency of oxidative phosphorylation that is conveyed by the serum. Relating other work in the field to our findings, we've added the following to our discussion: "Previous work in the field demonstrates that synchronization of the circadian clock in culture results in cycling of mitochondrial respiratory activity^{5,6} further underscoring the different effects of old serum, which does not support oscillations of oxidative phosphorylation associated transcripts. Age-dependent decrease in oxidative phosphorylation and increase in mitochondrial dysfunction⁷ has been seen in aged fibroblasts⁸ and contributes to age-related diseases⁹. We suggest that the age-related inefficiency of oxidative phosphorylation is conferred by serum signals to the cells such that oxidative phosphorylation cycles are mitigated. On the other hand, loss of cycling could contribute to impairments in mitochondrial function with age."

The delayed shifts after aged serum for clock transcripts (but not for Bmal1) are interesting and indicate that there may be a decoupling of Bmal1 transcript levels from the other clock gene phases. How do the authors interpret this? could it be related to altered chronotypes in the elderly?

One possible explanation is that the delay of NPAS2, BMAL1's binding partner, results in the delay of the transcription of clock controlled genes/negative arm genes. Since the RORs do not seem to be affected, Bmal is transcribed/translated as usual, but there isn't enough NPAS2 to bind with BMAL1. In this case downstream genes are slower to transcribe causing the phase delay.

Reviewer #2 (Public Review):

Schwarz et al. have presented a study aiming to investigate whether circulating factors in sera of subjects are able to synchronize depending on age, circadian rhythms of fibroblast. The authors used human serum taken from either old (age 70-76) or young (age 25-30) individuals to synchronise cultured fibroblasts containing a clock gene promoter driven luciferase reporter, followed by RNA sequencing to investigate whole gene expression.

This study has the potential to be very interesting, as evidence of circulating factors in sera that mediate peripheral rhythms has long been sought after. Moreover, the possibility that those factors are affected by age which could contribute to the weakened circadian rhythmicity observed with aging.

Here, the authors concluded that both old and young sera are equally competent at driving robust 24 hour oscillations, in particular for clock genes, although the cycling behaviour and nature of different genes is altered between the two groups, which is attributed to the age of the individuals. This conclusion could however be influenced by individual variabilities within and between the two age groups. The groups are relatively small, only four individual two females and two males, per group. And in addition, factors such as food intake and exercise prior to blood drawn, or/and chronotype, known to affect systemic signals, are not taken into consideration. As seen in figure 4, traces from different individuals vary heavily in terms of their patterns, which is not addressed in the text. Only analysing the summary average curve of the entire group may be masking the true data. More focus should be attributed to investigating the effects of serum from each individual and observing common patterns. Additionally, there are many potential causes of variability, instead or in addition to age, that may be contributing to the variation both, between the groups and between individuals within groups. All of this should be addressed by the authors and commented appropriately in the text.

We are not aware of any specific feature distinguishing the subjects (other than age) that could account for the differences between old and young. The fact that we see significant differences between the two groups, even with the relatively small size of the groups, suggests strongly that these differences are largely due to age. Nevertheless, we acknowledge that individual variability can be a contributing factor. For instance, the change in phase of clock genes appears to be driven largely by two subjects. We have commented on this and individual differences, in general, in the discussion.

The authors also note in the introduction that rhythms in different peripheral tissues vary in different ways with age, however the entire study is performed on only fibroblast, classified as peripheral tissue by the authors. It would be very interesting to investigate if the observed changes in fibroblast are extended or not to other cell lines from diverse organ origin. This could provide information about whether circulating circadian synchronising factors could exert their function systemically or on specific tissues. At the very least, this hypothesis should be addressed within the discussion.

It is likely that factors circulating in serum act on several tissues, and so their effects are relatively broad. However, this would require extensive investigation of other tissues. We now discuss this in the manuscript.

In addition to the limitations indicated above I consider that the data of the study is an insufficiently analysis beyond the rhythmicity analysis. Results from the STRING and IPA analysis were merely descriptive and a more comprehensive bioinformatic analysis would provide additional information about potential molecular mechanism explaining the differential gene expression. For example, enrichment of transcription factors binding sites in those genes with different patterns to pinpoint chromatin regulatory pathways.

We performed LinC similarity analysis (LISA) to study enrichment of transcription factor binding. Results are displayed in Fig 3B and in lines 157-168.

Recommendations for the authors:

The two reviewers and reviewing editor have agreed on the following recommendations for the authors:

Major:

(1) The bioinformatic analysis would benefit from a more thorough focus on variability between individuals. Specifically, the main conclusion of the manuscript could be significantly influenced by individual variabilities within and between the two age groups. This is of particular concern, as the groups are relatively small (four individual two females and two males, per group). In addition, the consideration of factors such as food intake and exercise prior to blood drawn, or/and chronotype, known to affect systemic signals should be more adequately explained. The lab is an experienced chronobiology lab, and thus we are confident that these factors had been thought of, but this needs to be better made clear.

As seen in Figure 4, traces from different individuals vary heavily in terms of their patterns, which is not addressed in the text. Only analysing the summary average curve of the entire group may be masking the relevant data. Furthermore, there are many potential causes of variability, instead or in addition to age, that may be contributing to the variation both, between the groups and between individuals within groups. All of this should be addressed by the authors and commented appropriately in the text.

We are not aware of any specific feature distinguishing the subjects (other than age) that could account for the differences between old and young. The fact that we see significant differences between the two groups, even with the relatively small size of the groups, suggests strongly that these differences are largely due to age. Nevertheless, we acknowledge that individual variability can be a contributing factor. For instance, the change in phase of clock genes appears to be driven largely by two subjects. We have commented on this and individual differences, in general, in the discussion.

(2) The study would benefit from a more thorough analysis of the data beyond the rhythmicity analysis. Results from the STRING and IPA analysis were merely descriptive and a more comprehensive bioinformatic analysis would provide additional information about potential molecular mechanism explaining the differential gene expression. For example, enrichment of transcription factors binding sites in those genes with different patterns to pinpoint chromatin regulatory pathways. This would provide additional value to the study, especially given the otherwise apparent lack of any mechanistic explanation.

We performed LinC similarity analysis (LISA) to study enrichment of transcription factor binding. Results are displayed in Fig 3B and in lines 157-168.

(3) There were some questions about the amplitude of the core circadian clock gene rhythms raised, which in other human cell types would be much higher. A comment on this matter and the provision of the raw luminescence traces for Fig 2A would be greatly beneficial.

Addressing the same topic: what are the typical fold changes of the many genes that change their rhythms after stimulation with young and old sera? For example, it would be useful to show histograms for the two groups. Does one group tend to have transcript rhythms of higher or lower fold changes? The presentation of the manuscript would further benefit from showing a few key examples for different types of responses.

The average luminescence trace for each individual serum sample from Fig 2A has been added to Fig S3A.

We've presented the fold change data in Figure S5. There are a few significant differences, but largely the groups are similar in terms of fold change.

(4) *There are several points that we recommend to consider to add to the discussion:*

What was the rationale to use these cells over the more common U2OS cells? Are there similarities between the rhythmic transcriptomes of the BJ-5TA cells and that of U2OS cells or other human cells? It should be relatively easy to address this point by assessing published datasets.

The original rationale to use BJ-5TA fibroblast cells was that we were aiming to build upon an observation found in a previous study² which showed that circadian period changes with age in human fibroblasts. While our findings did not match theirs, we think an added benefit of using the BJ-5TA line is that unlike U2OS cells, it is not carcinoma derived cell line. We've added this point in lines 98-101.

Our study finds many more rhythmic transcripts compared to the previous studies examining U2OS cells. This can be attributed to several factors including differences in methods, including the use of human serum in our study, cell type differences, or decoupling of rhythms in some cancer cells. While a comparison of BJ-5TA cells and U2OS cells could be interesting, a proper comparison requires investigation of many data sets, since any pair of BJ-5TA and U2OS data sets will most likely differ in some detail of experimental design or data processing pipeline, which could contribute to observed differences in rhythmic transcripts.

That being said, we compared clock reference genes (see Author response image 1) between BJ-5TA and U2OS cells, comparing circadian profiles obtained from our data with those available on CircaDB. These circadian profiles exhibit many similarities and a few differences. The peak to trough ratios (amplitudes) are quite similar for ARNTL, NR1D1, NR1D2, PER2, PER3, and are about 25% lower for CRY1 and somewhat higher for TEF (about 15%) in our data. We find that the MESORS are generally similar with the exception of NR1D1 which is much lower and NR1D2 which is much higher in our data.

For the rhythmic cell cycle genes, could this be the consequence of the serum which synchronizes also the cell cycle, or is it rather an effect of the circadian oscillator driving rhythms of cell cycle genes?

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It may not be the cycling *per se*, but rather an overall inefficiency of oxidative phosphorylation that is conveyed by the serum. Relating other work in the field to our findings, we've added the following to our discussion: "Previous work in the field demonstrates that synchronization of the circadian clock in culture results in cycling of mitochondrial respiratory activity^{5,6} further underscoring the different effects of old serum, which does not support oscillations of oxidative phosphorylation associated transcripts. Age-dependent decrease in oxidative phosphorylation and increase in mitochondrial

dysfunction⁷ is seen also in aged fibroblasts⁸ and contributes to age-related diseases⁹. We suggest that the age-related inefficiency of oxidative phosphorylation is conferred by serum signals to the cells such that oxidative phosphorylation cycles are mitigated. On the other hand, loss of cycling could contribute to impairments in mitochondrial function with age.”

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One possible explanation is that the delay of NPAS2, BMAL1’s binding partner, results in the delay of the transcription of clock controlled genes/negative arm genes. Since the RORs do not seem to be affected, Bmal is transcribed/translated as usual, but there isn’t enough NPAS2 to bind with BMAL1. In this case downstream genes are slower to transcribe causing the phase delay.

The discussion would also benefit from mentioning parallels and dissimilarities with previous works, as well as what would be possible mechanisms for such an effect.

We’ve expanded our discussion in the manuscript to discuss possible mechanisms and also how the genes/pathways implicated in our study relate to other aging literature.

Minor:

While time of serum collection is provided in the methods, it would be very useful to provide this information, along with the accompanying argumentation also at a more prominent position and to also add it to Table S1.

We made sure to highlight the collection time in the abstract of the manuscript “We collected blood from apparently healthy young (age 25-30) and old (age 70-76) individuals at 14:001 and used the serum to synchronize cultured fibroblasts.” The time of blood draw is also in sections of the paper (Intro and Methods). Since Table S1 is demographic information, we did not think that the blood draw time fit best there, but hopefully it is now clear in the text.

L73 EKG: define the abbreviation

We rewrote this paragraph, but defined the term where it is used the paper.

L77: transfected BJ-5TA fibroblasts. Mention in the text that these are stably transfected cells.

We added this to the text.

L88: Day 2 also revealed different phases of cyclic expression between young and old "groups" for a larger number of genes. Here it is only two donors, right?

Yes, we swapped out the word “groups” for “subjects”.

L115. MESORs of steroid biosynthesis genes, particularly those relating to cholesterol biosynthesis, were also increased in the old sera condition. This is quite interesting, can the authors speculate on the significance of this finding?

We’ve added discussion about this finding in the context of the literature in our discussion.

Fig 3. - FDRs are only listed for certain KEGG pathways, and gene counts for each pathway are also missing, which excludes some valuable context for drawing conclusions. Full tables of KEGG pathway enrichment outputs should be provided in supplementary materials. Input gene lists should also be uploaded as supplementary data files.

Both output and input files are included in this submission as additional files.

Line 322 - How many replicates were excluded in the end for each group? Providing this information would strengthen the claim that the ability of both old and young serum to drive 24h oscillations in fibroblasts is robust and not only individual.

Each serum was tested in triplicate in two individual runs of the experiment. Of the 15 serum samples, on one of the runs, a triplicate for each of two serum samples (one old, one young) was excluded. Given that only one technical replicate in one run of the experiment had to be excluded for one old and one young individual out of all the samples assayed, this supports the idea that young and old serum drive robust oscillations.

Line 373 - Should list which active interaction sources were used for analysis.

In this manuscript we used STRING (search tool for retrieval of interacting genes) analysis to broadly identify relevant pathways defined by different algorithms. From these data, we focused in particular on KEGG pathways.

Reviewer #1 (Recommendations For The Authors):

These comments are in addition to those provided above:

Minor:

L73 EKG: define the abbreviation

We rewrote this paragraph, but defined the term where it is used the paper.

L77: transfected BJ-5TA fibroblasts. Mention in the text that these are stably transfected cells.

We added this to the text.

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Yes, we swapped out the word "groups" for "subjects".

L115. MESORs of steroid biosynthesis genes, particularly those relating to cholesterol biosynthesis, were also increased in the old sera condition. This is quite interesting, can the authors speculate on the significance of this finding?

We've added discussion about this finding in the context of the literature.

Fig.4 The fold change amplitude of the clock gene seems quite a bit lower than what is usually expected (for Nr1d1 it is usually 10 fold). The authors should provide an explanation and discuss this.

There are a variety of factors that contribute to the fold change amplitude of clock genes. First, the change in amplitude of clock genes is lower *in vitro* compared to *in vivo* samples. For example, in U2OS cell cultures the fold change in the cycling of Nr1d1 is only 2 fold and is not significantly different from the fold change we observe (as shown in the U2OS data from CircaDB plotted in Figure 1R). Second, the method of synchronization contributes to the strength of the rhythms. Serum synchronization is generally less effective at driving strong clock cycling than forskolin or dexamethasone although, as noted in the manuscript, it may promote the cycling of more genes. Lastly, rhythm amplitude is also dependent on the cell type in question so cell to cell variability also contributes to differences. However, overall, we do not find major differences in comparing the U2OS data and ours. Please note that the y-axis has a logarithmic scale.

What is the authors' strategy to identify which serum components that are responsible for the reported changes? This should be discussed.

In the future, we intend to analyze the serum factors using a combination of fractionation and either proteomics or metabolomics to identify relevant factors. We have added this to the discussion.

Reviewer #2 (Recommendations For The Authors):

Overall, the article is well-written but lacks some more rigorous data analysis as mentioned in the public review above. In addition to a more thorough analysis approach focusing much more heavily on individual variability, several other changes can be made to strengthen this study:

Fig 3. - FDRs are only listed for certain KEGG pathways, and gene counts for each pathway are also missing, which excludes some valuable context for drawing conclusions. Full tables of KEGG pathway enrichment outputs should be provided in supplementary materials. Input gene lists should also be uploaded as supplementary data files.

Both output and input files are included in this submission as additional files.

Fig 1A. - Only n=5 participants were used for this analysis, explanation of the exclusion criteria for the other participants would be useful.

As Figure 1A is a schematic, we assume the reviewer is referring to Figure 1B. We've provided a flow chart of subject inclusion/exclusion in Figure S2.

Fig 2. - For circadian transcriptome analysis only n=4 participants were used - what criteria was used to exclude individuals, and why were only these individuals used in the end?

As patient recruitment was interrupted by COVID, we selected samples where we had sufficient serum to effectively carry out the RNA seq experiment and control for age and sex.

Line 322 - How many replicates were excluded in the end for each group? Providing this information would strengthen the claim that the ability of both old and young serum to drive 24h oscillations in fibroblasts is robust and not only individual.

Each serum was tested in triplicate in two individual runs of the experiment. Of the 15 serum samples, on one of the runs, a triplicate for each of two serum samples (one old, one young) was excluded. Given that only one technical replicate in one run of the experiment had to be

excluded for one old and one young individual out of all the samples assayed, this supports the idea that young and old serum drive robust oscillations.

| Line 373 - Should list which active interaction sources were used for analysis.

In this manuscript we used STRING (search tool for retrieval of interacting genes) analysis to identify relevant pathways. We do not present any STRING networks in the paper.

| Line 68 - "These novel findings suggest that it may be possible to treat impaired circadian physiology and the associated disease risks by targeting blood borne factors." This is a completed overstatement that are cannot be sustained by the limited findings provided by the authors.

We've modified this statement to avoid overstating results.

- (1) Pagani, L. *et al.* Serum factors in older individuals change cellular clock properties. *Proceedings of the National Academy of Sciences* **108**, 7218–7223 (2011).
- (2) Pagani, L. *et al.* Serum factors in older individuals change cellular clock properties. *Proc Natl Acad Sci U S A* **108**, 7218–7223 (2011).
- (3) Lee, Y. *et al.* G1/S cell cycle regulators mediate effects of circadian dysregulation on tumor growth and provide targets for timed anticancer treatment. *PLOS Biology* **17**, e3000228 (2019).
- (4) Tomasetti, C. *et al.* Cell division rates decrease with age, providing a potential explanation for the age-dependent deceleration in cancer incidence. *Proceedings of the National Academy of Sciences* **116**, 20482–20488 (2019).
- (5) Cela, O. *et al.* Clock genes-dependent acetylation of complex I sets rhythmic activity of mitochondrial OxPhos. *Biochimica et Biophysica Acta (BBA) - Molecular Cell Research* **1863**, 596–606 (2016).
- (6) Scrima, R. *et al.* Mitochondrial calcium drives clock gene-dependent activation of pyruvate dehydrogenase and of oxidative phosphorylation. *Biochimica et Biophysica Acta (BBA) - Molecular Cell Research* **1867**, 118815 (2020).
- (7) Lesnefsky, E. J. & Hoppel, C. L. Oxidative phosphorylation and aging. *Ageing Research Reviews* **5**, 402–433 (2006).
- (8) Greco, M. *et al.* Marked aging-related decline in efficiency of oxidative phosphorylation in human skin fibroblasts. *The FASEB Journal* **17**, 1706–1708 (2003).
- (9) Federico, A. *et al.* Mitochondria, oxidative stress and neurodegeneration. *Journal of the Neurological Sciences* **322**, 254–262 (2012).

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